

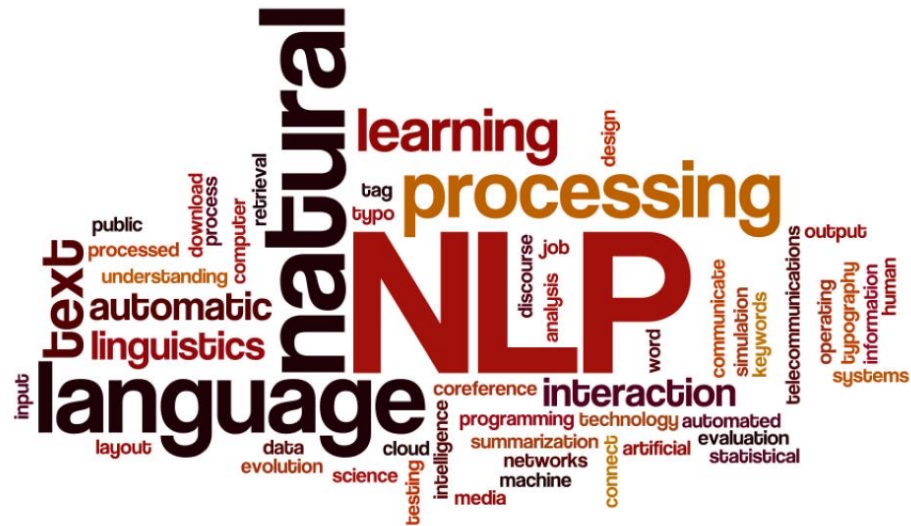
Class 4

CSX4109/INX4109 AI for Business

NATURAL LANGUAGE PROCESSING

SEMESTER 1/2023





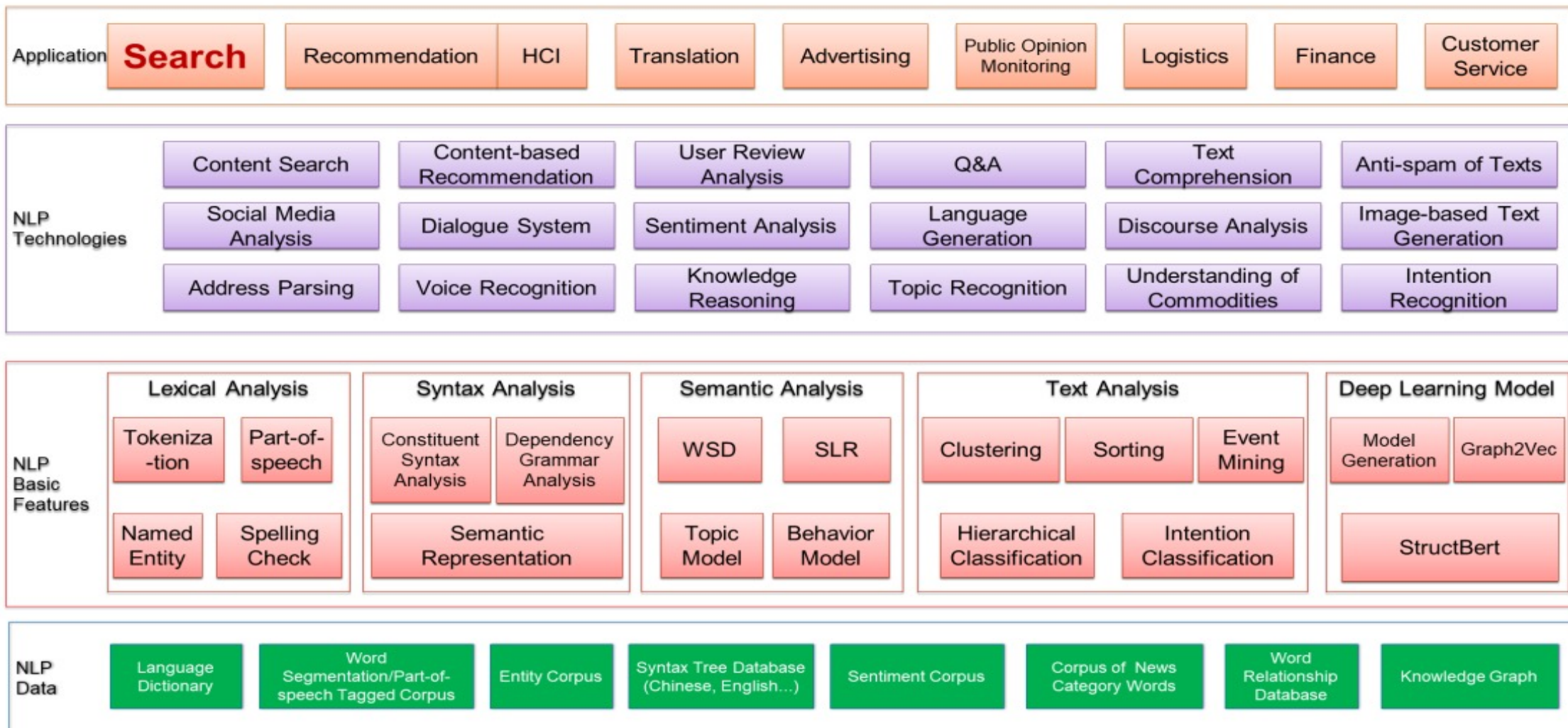
What is NLP?

Natural Language Processing (NLP) is the ability to **receive** language information, **interpret** it, **complete a task** based on the language information and **produce** appropriate human language.
[John Medicine, 2020]

How is NLP used in these applications?



Capabilities of NLP



Areas of NLP



How to Read

- Natural Language Understanding



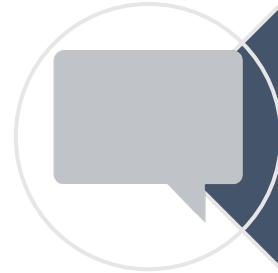
How to Write

- Natural Language Generation

Formats of Inputs

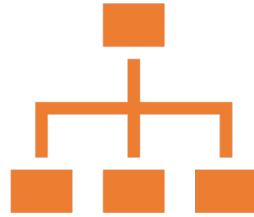


Text



Speech

Analysis in **NLP**



Syntactic Analysis

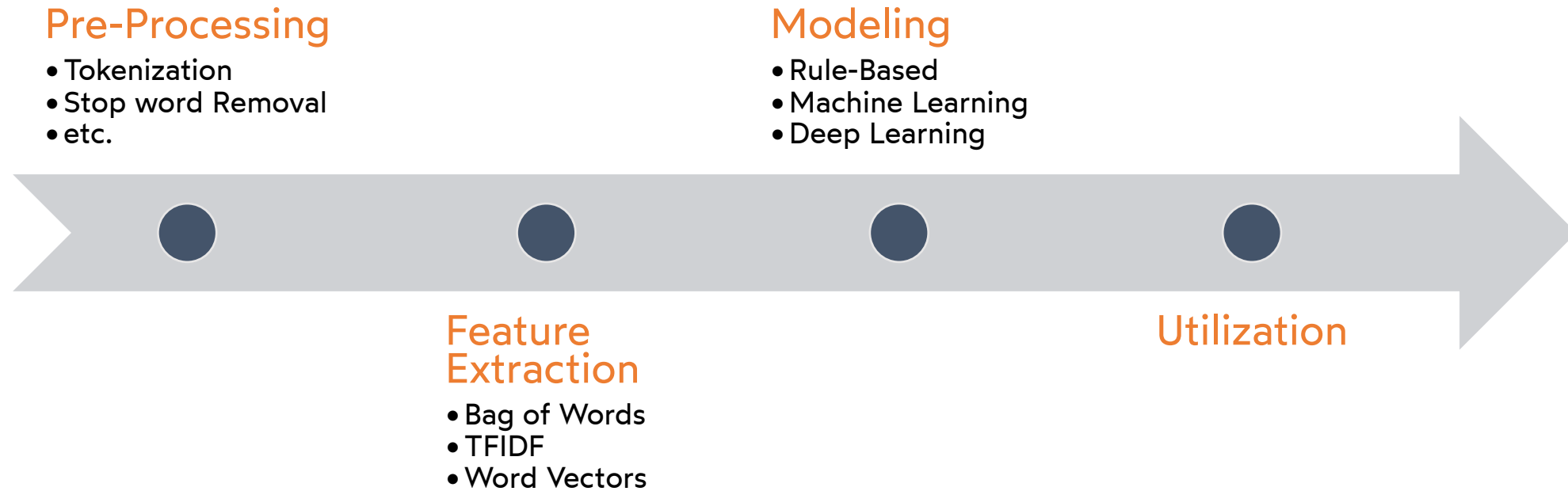
Structure



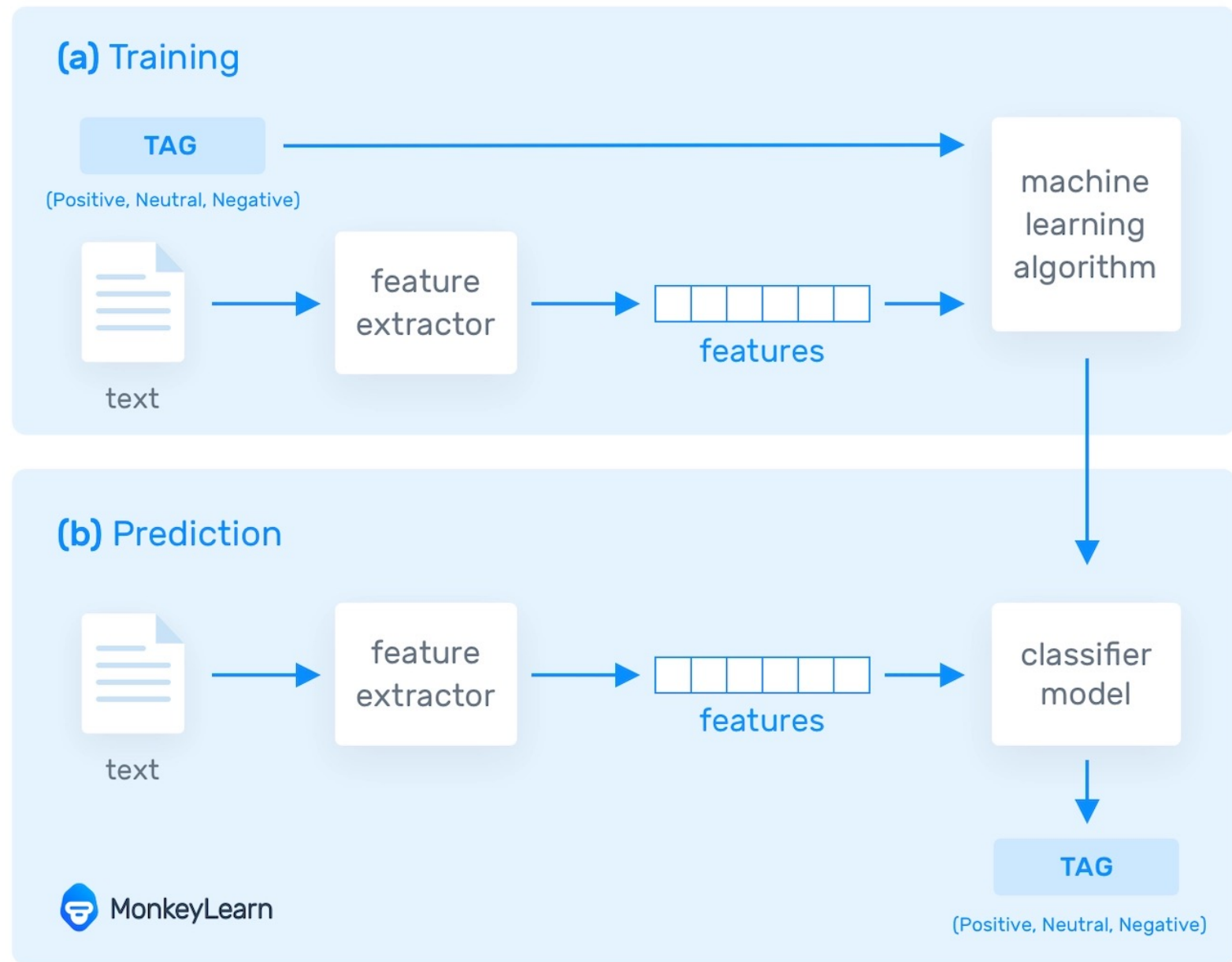
Semantic Analysis

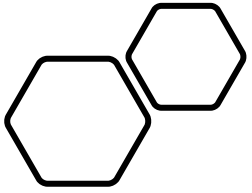
Meaning

Steps in NLP



ML Process in NLP





Preparing/cleaning the data in
such a way that it can be fed to
the model.

Pre-Processing

Pre-Processing

- Tokenization
- Stop Words Removal
- Normalization
 - Stemming
 - Lemmatization
- Spelling Correction

Tokenization

- The process of separating streams of text into smaller units (which can be sentences, tokens or words).
- **Sentence tokenization** breaks long text into sentences.
- **Word tokenization** splits text in a sentence into words.
- **Language Issues:**
 - No word boundaries (no space between words) e.g. Chinese, Japanese, Thai
 - No sentence boundaries (no period at the end of the sentence) e.g. Thai
- **Challenges:**
 - Handling words that are often appearing together.
 - Artificial Intelligence, Computer Science, Machine Learning

Example: Tokenization

- I love to travel.

I love to travel

- Artificial Intelligence can make it better.

Artificial Intelligence can make it better

Artificial Intelligence can make it better

Stop Words Removal

- Stop words are **commonly used** words in the language.
 - a, an, the, in, on, and etc.
- They are normally considered **unimportant**; hence, usually are removed so the algorithm can focus on the more important ones.
- The set of stop words to use and the decision whether to remove them depend on applications.

I need help on how to adopt AI in my business.



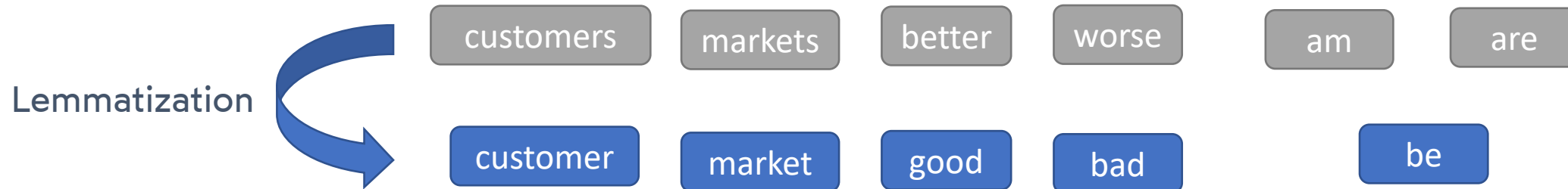
need help how adopt AI business

i, in, on, my, to

Word Normalization

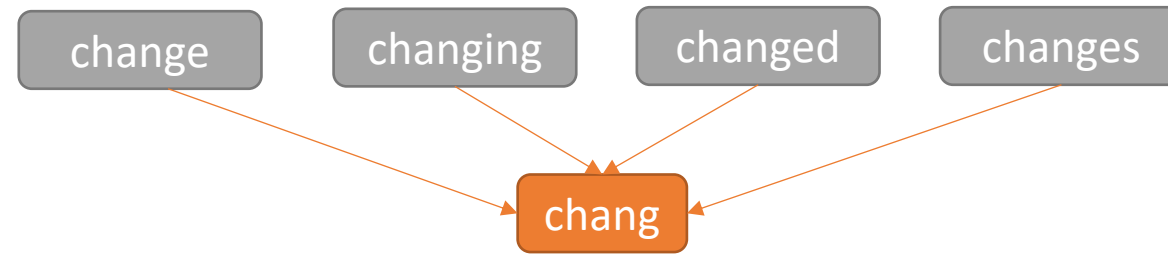
- Both lemmatization and stemming reduce the inflected word to its root form.
- **Stemming**
 - is based on heuristic algorithm that removes the end of words.
 - may not give out the actual language words.
- **Lemmatization**
 - uses morphological analysis of words.
 - returns the dictionary form of the words (*aka* lemma).

Example: Stemming vs. Lemmatization

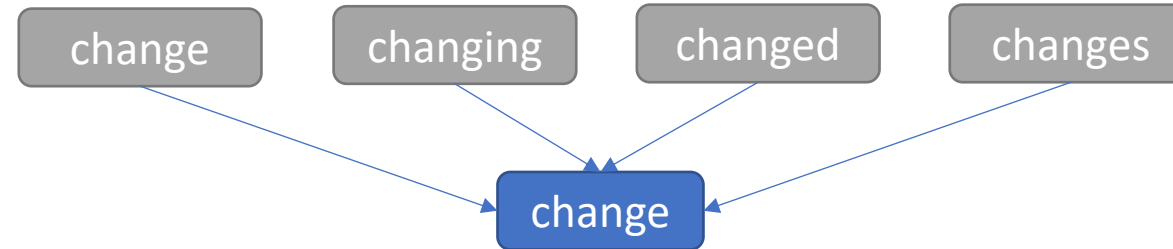


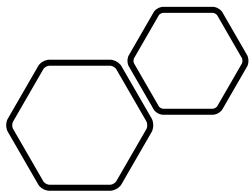
Example: Stemming vs. Lemmatization

Stemming



Lemmatization





NLP Subtasks

Part-of-Speech Tagging

- *aka* POS Tagging
- The process of assigning a **part-of-speech** or lexical class marker to each word in a collection.
- **Traditional** Parts of Speech
 - noun, verb, adjective, preposition, adverb, article, interjection, pronoun, conjunction, etc.

The	suite	was	very	good	with	comfortable	furniture
DET	N	V	ADV	ADJ	P	ADJ	N

Named Entity Recognition

- The process of detecting named entities in the given text.
- Useful in co-reference resolution.
- Examples of named entities:
 - Person's name e.g. James Bond, Joe Biden
 - Location e.g. Bangkok, New York City
 - Organization name e.g. Assumption University, Bank of America

Joe Biden is the current president of the United States of America.



Joe Biden is the current president of the United States of America.

Person

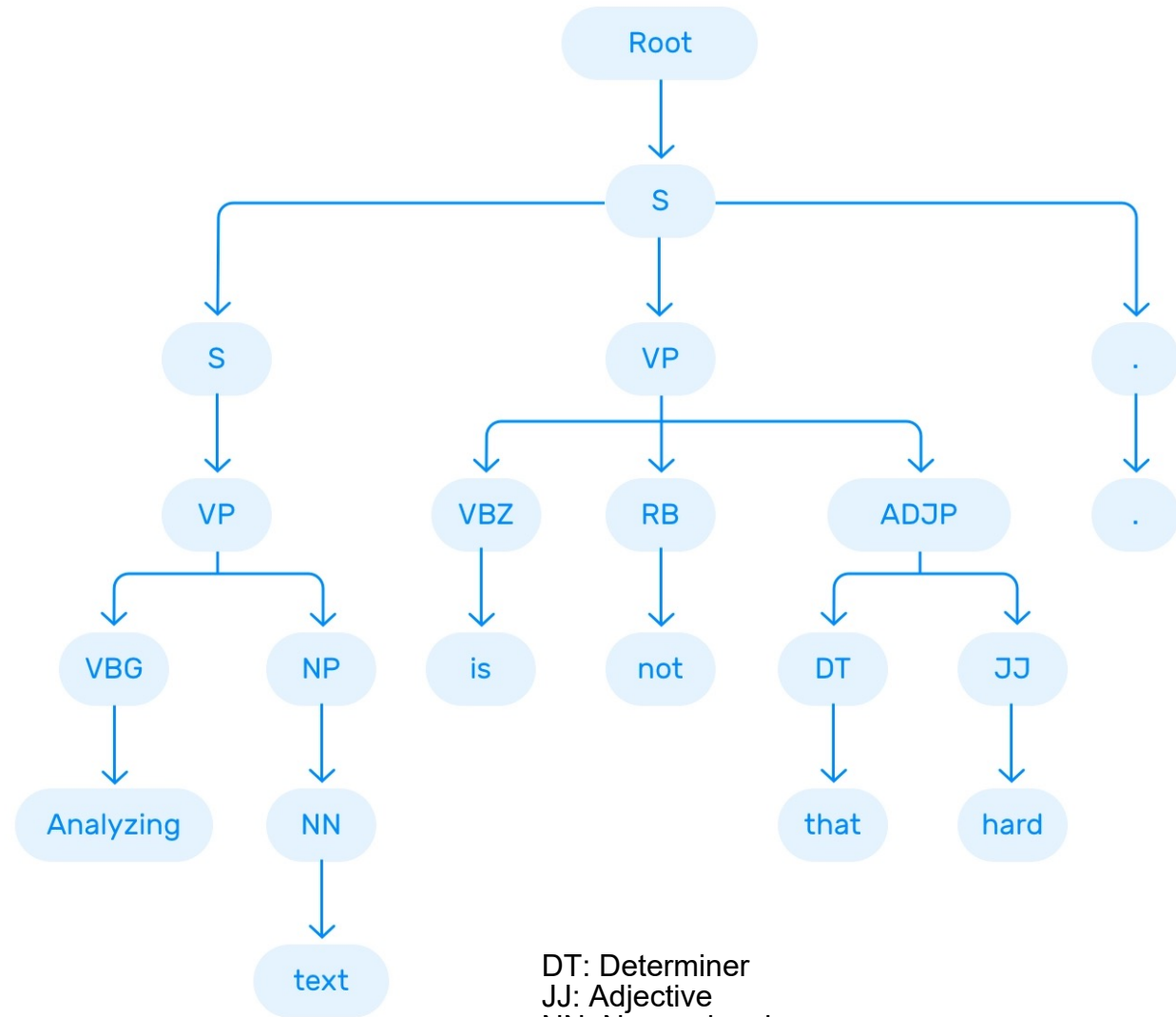
Country

Parsing

Understanding the structure of the text, check if it conforms to the grammar and form a parse tree.

Constituency Parsing

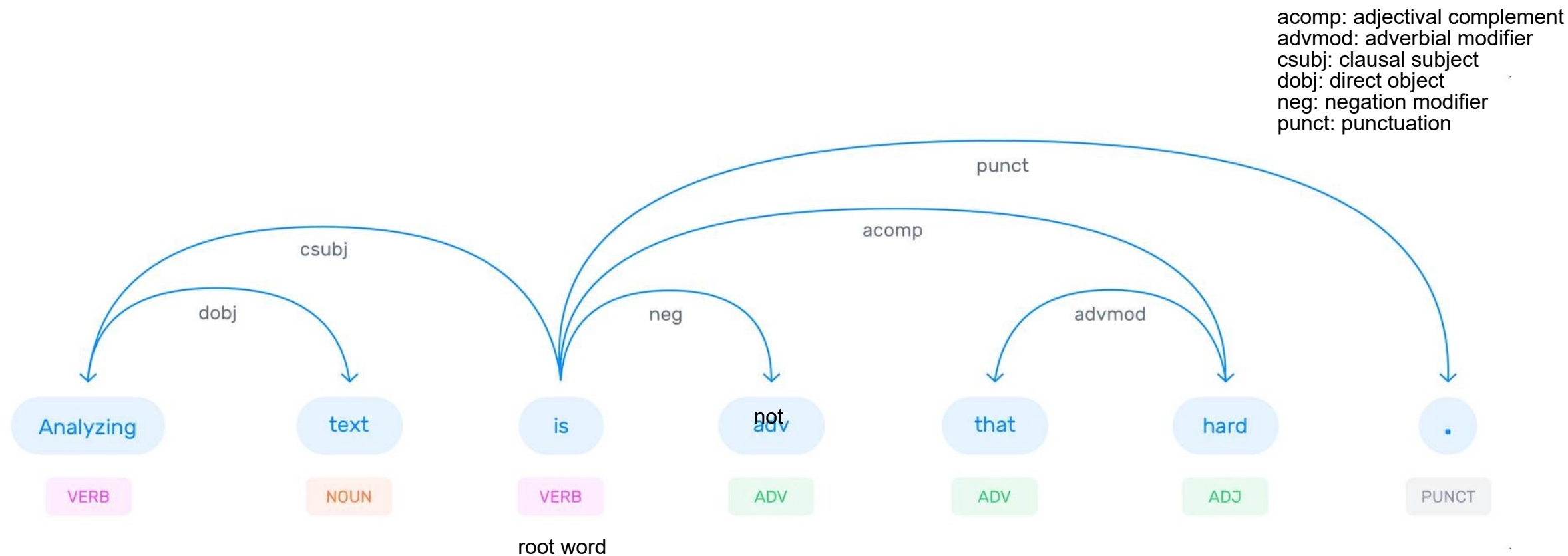
Analyze the **syntactic structure** of a sentence (phrase structure).



DT: Determiner
JJ: Adjective
NN: Noun, singular or mass
RB: Adverb
S: Simple declarative clause
VBG: Verb, gerund or present participle
VBZ: Verb, 3rd person singular present

Dependency Parsing

Analyze how words in a sentence are **related**.



Word Sense Disambiguation

- Natural language is ambiguous.
- Some words have different meaning in different context.

John went to the bank.

Financial institution?

Bank of the river?

Put the box on the table in the kitchen

Is the box already on the table, and
to be put in the kitchen

Is the box to be put on the table which
is in the kitchen?

Tasks of NLP



NLP TASKS [4]

SPEECH RECOGNITION

- Speech-to-text
- Required for application follows voice commands or answer spoken question

SPEECH TAGGING

- Grammatical tagging
- Determine a part of speech, word or text-based on its context

WORD SENSE DISAMBIGUATION

- Semantic analysis
- What make the most sense in the given context

NAMED ENTITY RECOGNITION

- Co-reference resolution
- Identify words, phrases or idiom as useful entities

SENTIMENT ANALYSIS

Extract subjective qualities – attitudes, emotions, sarcasm, confusion, suspicion

NL GENERATION

Translate a structured information into human language

Tools of NLP

PYTHON AND THE NATURAL LANGUAGE TOOLKIT (NLTK)

Libraries for handling
specific NLP tasks such as:

- Sentence parsing
- Word segmentation
- Stemming and
lemmatization
- Tokenization

STATISTICAL NLP, MACHINE LEARNING, AND DEEP LEARNING

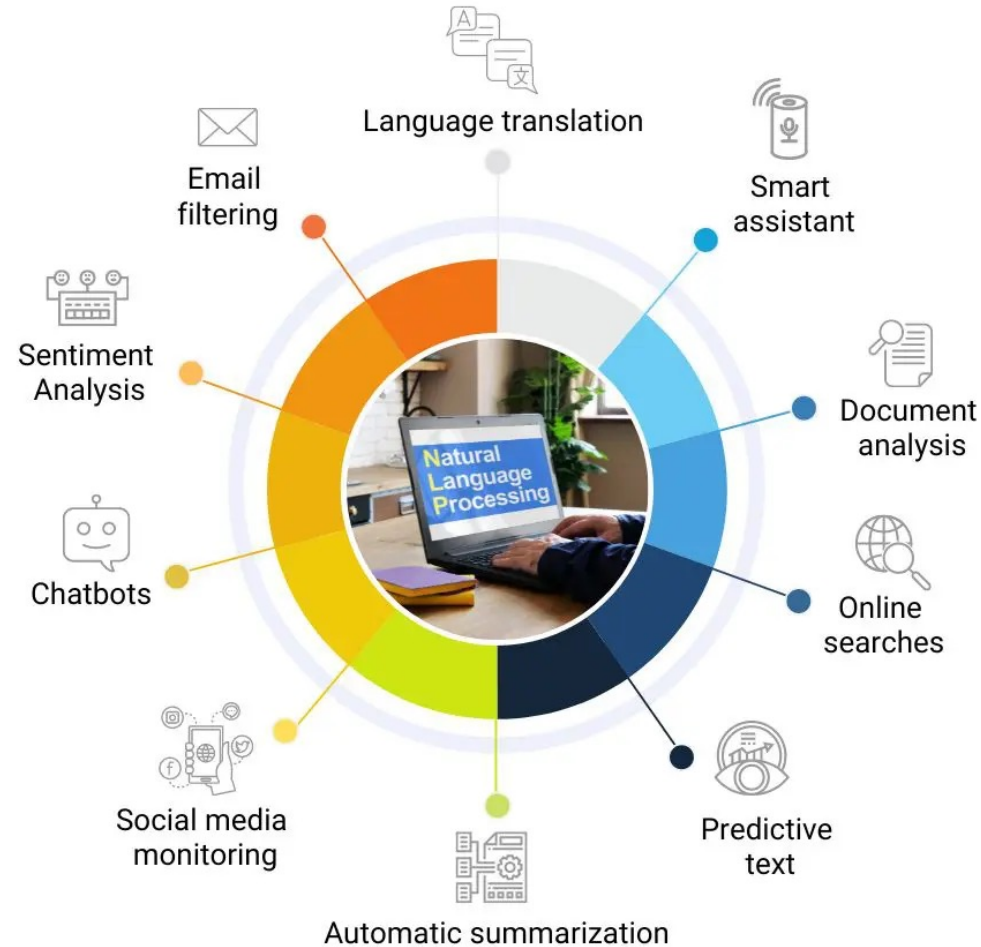
Combines computer algorithms with machine learning and deep learning models to automatically extract, classify, and label elements of text and voice data and then assign a statistical likelihood to each possible meaning of those elements.

COMMERCIAL NLP TOOLS

- IBM Watson
- Microsoft Cognitive Services, Azure HDInsight
- Google Cloud Auto ML, Natural Language API, Natural Language AI
- AWS Language Services

Applications of NLP

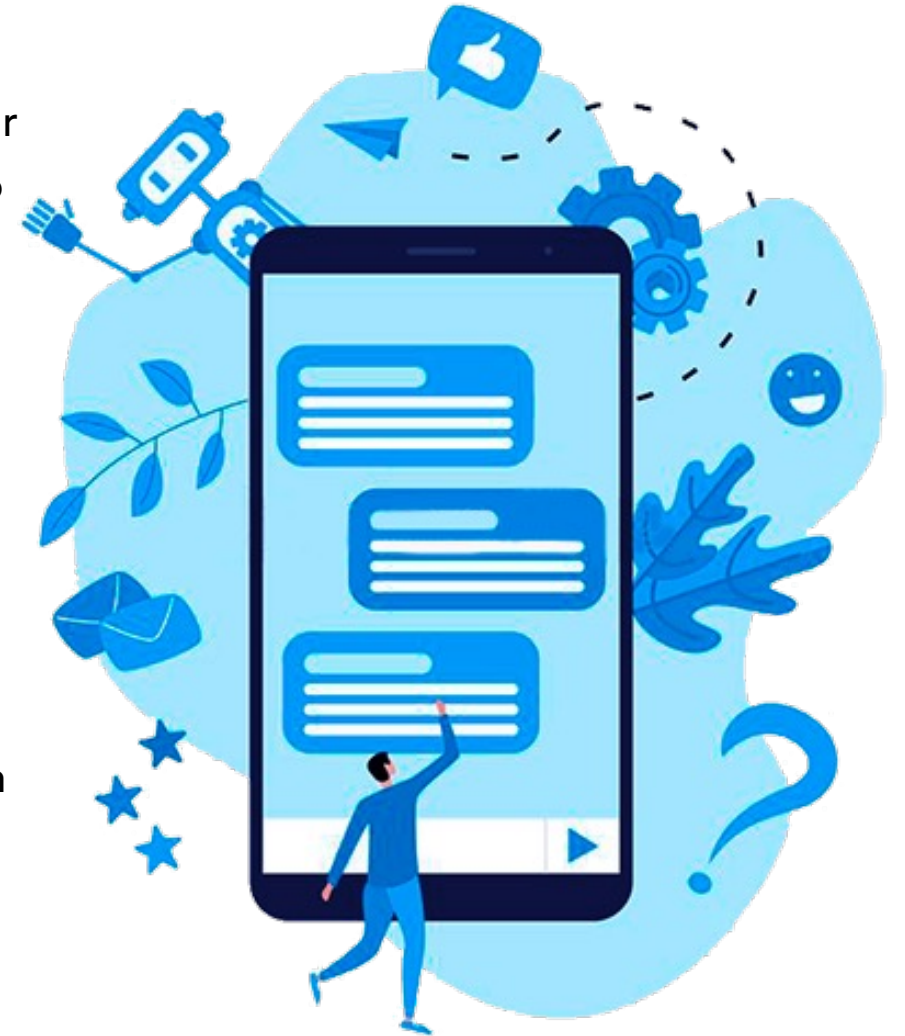
Applications of Natural Language Processing



<https://datasciencedojo.com/blog/natural-language-processing-applications/>

Applications of NLP

- As a personal development we consider improvement in **all areas of life** that do not belong to the business and professional life.
- These include **family and friend relations, personal interests, life balance** and the like.
- NLP helps us to improve **self-esteem, assertiveness and flexibility**.
- The most important area of application is learning how to set up and realize goals and recognizing the truly important things in our lives.



Applications of NLP

MACHINE TRANSLATION:

USE CASE

- Effective translation must accurately capture the **meaning** and **tone** of the input language and translate it to text with the same meaning and desired impact in the output language.

APPLICATION

- Google Translates

Applications of NLP

VIRTUAL AGENTS:

USE CASE

- The use speech recognition to recognize patterns in **voice commands** and natural language generation to respond with appropriate action or helpful comments.

APPLICATION

- Apple's Siri
- Google Assistant
- Amazon's Alexa
- Microsoft Cortana
- Samsung Bixby

VIRTUAL AGENTS (CONT.):

Applications of NLP



Applications of NLP

CHATBOTS:

USE CASE

- The **response to typed text** entries by learning to recognize contextual clues about human requests and use them to provide even better responses or options over time.

APPLICATION

- Line Messaging API
- Google Chatbot AI
- [Azure Bot Service](#)

Applications of NLP

SMART DEVICES:

USE CASE

- The use speech recognition to recognize patterns in voice commands and natural language generation to respond with appropriate action or helpful comments.
- The response to typed text entries by learning to recognize contextual clues about human requests and use them to provide even better responses or options over time.

APPLICATION

- Smart Watch



Applications of NLP

TEXT SUMMARIZATION:

USE CASE

- Text summarization uses NLP techniques to digest huge volumes of digital text and create summaries and synopses for indexes, research databases, or busy readers who don't have time to read full text.

APPLICATION

- [Simplify](#)
- [Summary Generator](#)
- [AppZaza Article Summarizer](#)
- [Summarizer](#)
- [Autosummarizer](#)
- [Text Compactor](#)
- [Resoomer](#)

Applications of NLP

EMAIL ASSISTANT & SPAM DETECTION:

USE CASE

- NL capability includes overuse of financial terms, characteristic bad grammar, threatening language, inappropriate urgency, misspelled company names, and more [2].
- The best spam detection technologies use NLP's text classification capabilities to scan emails for language that often indicates spam or phishing [3].

APPLICATION

- Email Assistants
- Spam Detection
- Fake news and Cyberbullying Detection

SOCIAL MEDIA SENTIMENT ANALYSIS:

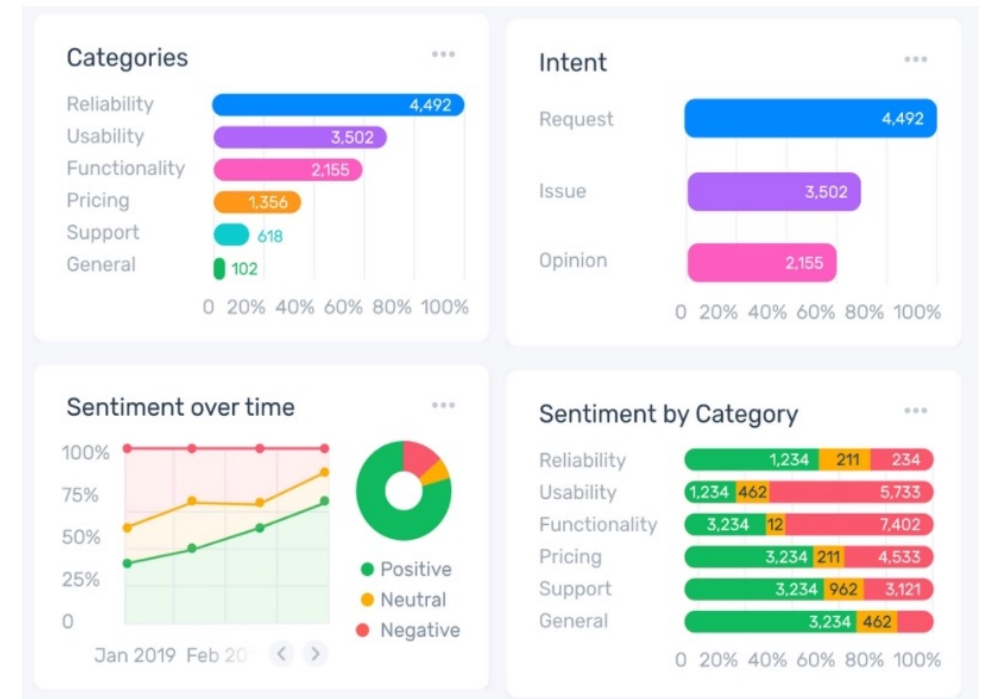
USE CASE

- An essential business tool for **uncovering hidden data insights** from social media channels.
- Sentiment analysis can analyze language used in social media posts, responses, reviews, and more to **extract attitudes and emotions** in response to products, promotions, and events—information companies can use in product designs, advertising campaigns, and more.

APPLICATION

- Sentimental Analysis Tools
- Social Listening / Monitoring Tools

Applications of NLP



CUSTOMER SERVICE AUTOMATION:

USE CASE

Applications of NLP

- NLP has excellent results in the development of public speaking skills, relationship management, conflict management and communication skills in the broadest sense.
- NLP can encourage development of creativity and can be of great use in strategic planning, and determination of mission, vision and values of the organization.

APPLICATION

- Text analytics to extract important trends from customer feedback in many forms.

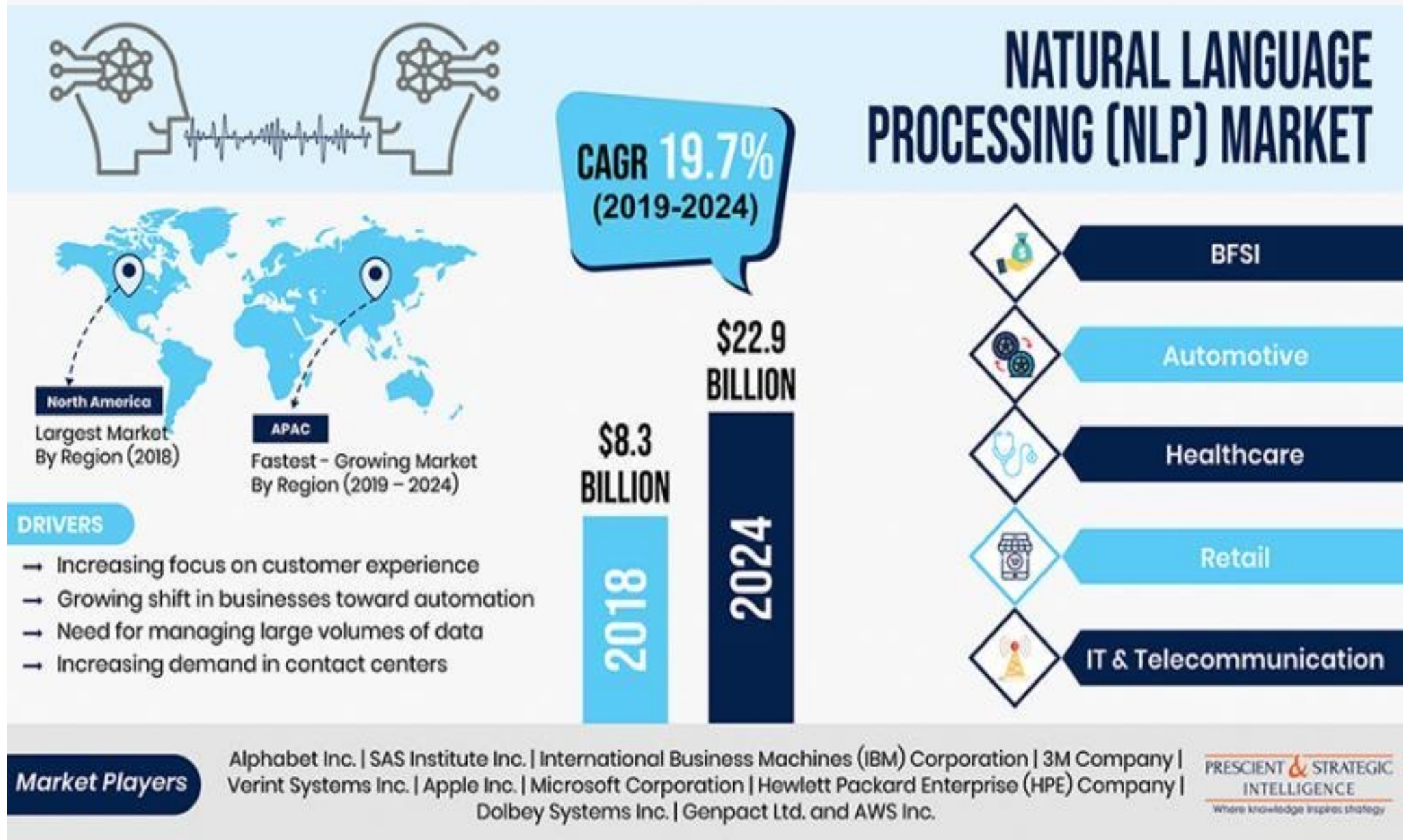


Applications of NLP

- NLP can help employees to quickly and efficiently adapt to different types of changes that today's business world is exposed to daily.

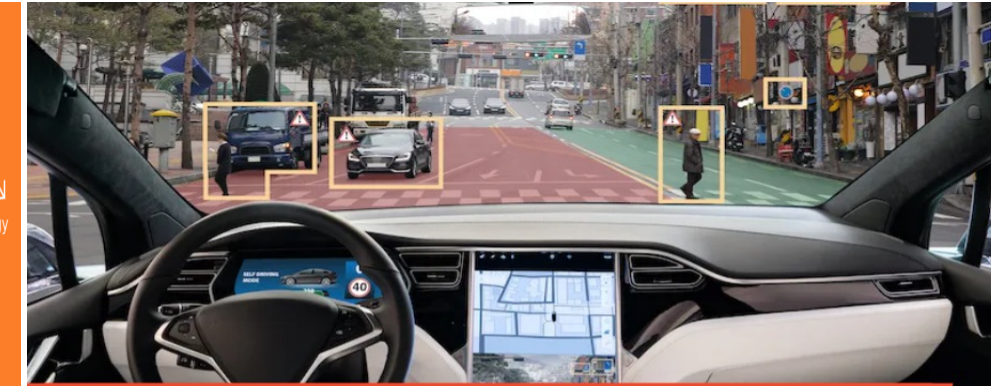
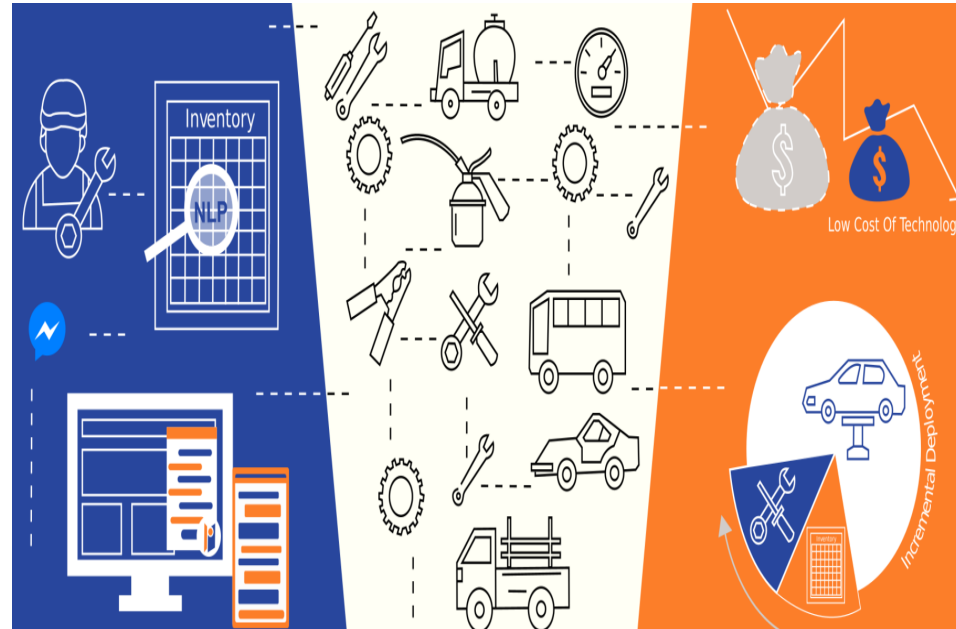


NLP Market



Business Application of NLP

NLP IN AUTOMOTIVE:



**AUTONOMOUS VEHICLES AND
NATURAL LANGUAGE
PROCESSING ON TRAVEL'S 2021**

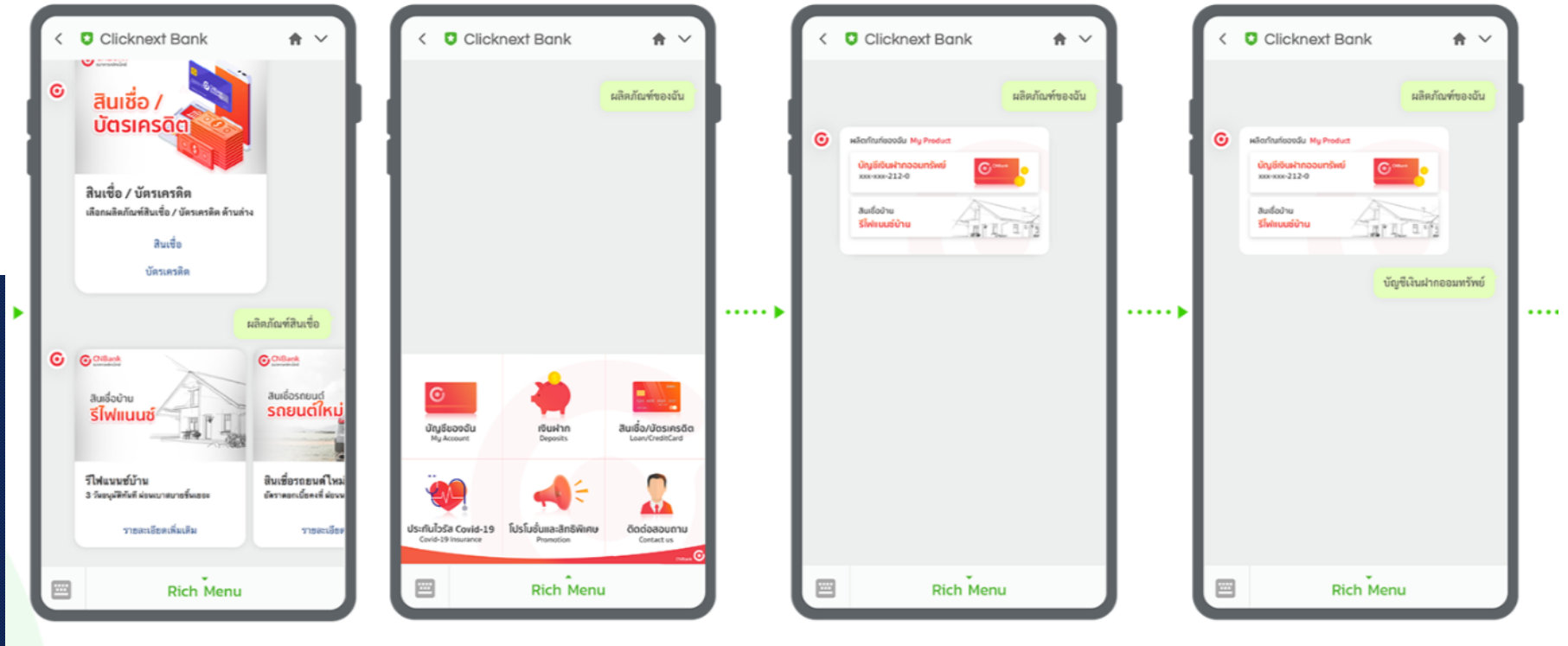
NATURAL LANGUAGE PROCESSING ACROSS THE AUTOMOTIVE VALUE CHAIN

A multi-threaded text analytics service, Natural Language Processing methods and hierarchic clustering of documents.

Text Analytics Service/ Document Classification

Business Application of NLP

NLP IN BANKING, FINANCIAL SERVICES AND INSURANCE (BFSI):



NLP IN BANKING, FINANCIAL SERVICES AND INSURANCE (BFSI):

DIGITAL FINANCIAL COACH / ADVISOR

Business Application of NLP

- NLP in the financial industry is being used to build **transactional bots**. These bots act as a digital financial advisor helping users deal with savings, investments and other financial plans. Services of this nature help with engagement and provide users with better user experience. Transactional bots are built using NLP technology to process data in the form of human language.
- A good example of NLP and ML in FinTech is **Cleo**, a personal financial assistant. Cleo can set personal budgets, provide financial advice and help clients meet their personal financial goals.
- Digital assistants are also being used in other scenarios such as dividend management, term life renewals, reminder of credit card payments and miscellaneous other notifications consumers generally need to keep a track of.

NLP IN BANKING, FINANCIAL SERVICES AND INSURANCE (BFSI):

CREDIT SCORING:

Business Application of NLP

- **Evaluating the creditworthiness** of an individual and **understanding the risk profile** is a vital challenge banks and insurance companies face regularly. With the advent of FinTech AI banking, advanced technologies like machine learning and NLP have contributed significantly in analyzing borrower data and facilitating assessing risks and creditworthiness of individuals.
- LenddoEFL, a Singapore-based company offers a software called LenddoScore that uses advanced NLP and machine learning algorithms to assess creditworthiness of borrowers. The platform combs through the prospective borrower's digital footprints across social media, browsing history, geo-location and other information on personal devices. The data is then converted into a credit score using machine learning algorithms that institutions can use to evaluate the creditworthiness of the individual.

NLP IN BANKING, FINANCIAL SERVICES AND INSURANCE (BFSI):

CUSTOMER SERVICE:

Business Application of NLP

- NLP is being incorporated across **chatbot** and other software to enhance customer service and **resolve queries**. With features like sentiment analysis, NLP is often considered a suitable technology for driving better results from customer services and support.
- We take the example of **Ocado**, an online supermarket based in the UK that has been using NLP and sentiment analysis to identify urgent queries. This has resulted in them replying to urgent queries 4x times faster bringing about efficiency and a happier lot of customers.
- Using NLP application the software identifies the most relevant information, while sentiment analysis helps the customer service teams to provide better answers to customer queries.

NLP IN BANKING, FINANCIAL SERVICES AND INSURANCE (BFSI):

CONTRACT ANALYSIS:

Business Application of NLP

- Contract analysis is an internal task that is repetitive, mundane and can easily be done with the help of NLP and machine learning. Enterprises like JP Morgan have had high success using NLP powered software to analyze contracts, helping them to **free up 360,000 man-hours, yearly**.
- In addition to using NLP in the banking industry, there has also been a rise in the use of related deep tech, such as Optical Character Recognition (OCR) to digitize hard copies of documents, scan to analyze and correct contracts etc. Similarly, software like Nuance Document Finance Solution Communication help finance companies automate and digitize their documentation using NLP.

NLP

in Industries

Natural Language Processing in Healthcare is expected to grow from USD 1030.2 million in 2016 to USD 2650.2 million in 2021, at a CAGR of 20.8 percent during the forecast period.

NLP IN HEALTHCARE:

01

Handle the Surge in Clinical Data



02

Support Value-Based Care & PHM Applications



03

Improve Patient-Provider Interactions with EHR



04

Empower Patients with Health Literacy



05

Address the Need for Higher Quality of Healthcare



06

Identify Patients who need Improved Care



Natural Language Processing in the Healthcare industry can help enhance the **accuracy** and **completeness** of **EHRs** by transforming the free text into standardized data. It could also make documentation easier by allowing care providers to dictate **notes** as NLP turns it into documented data.

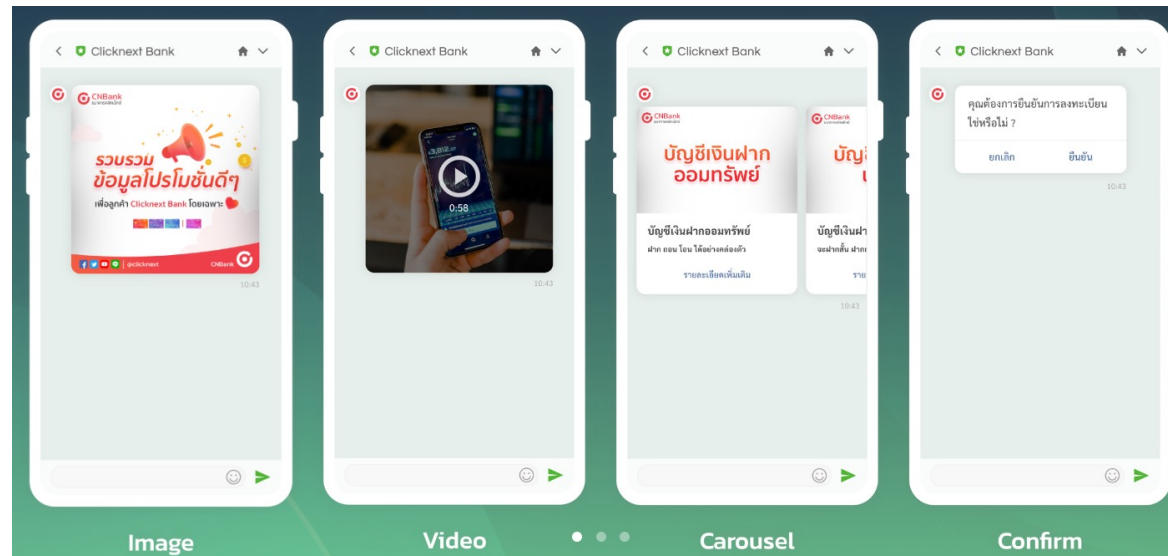
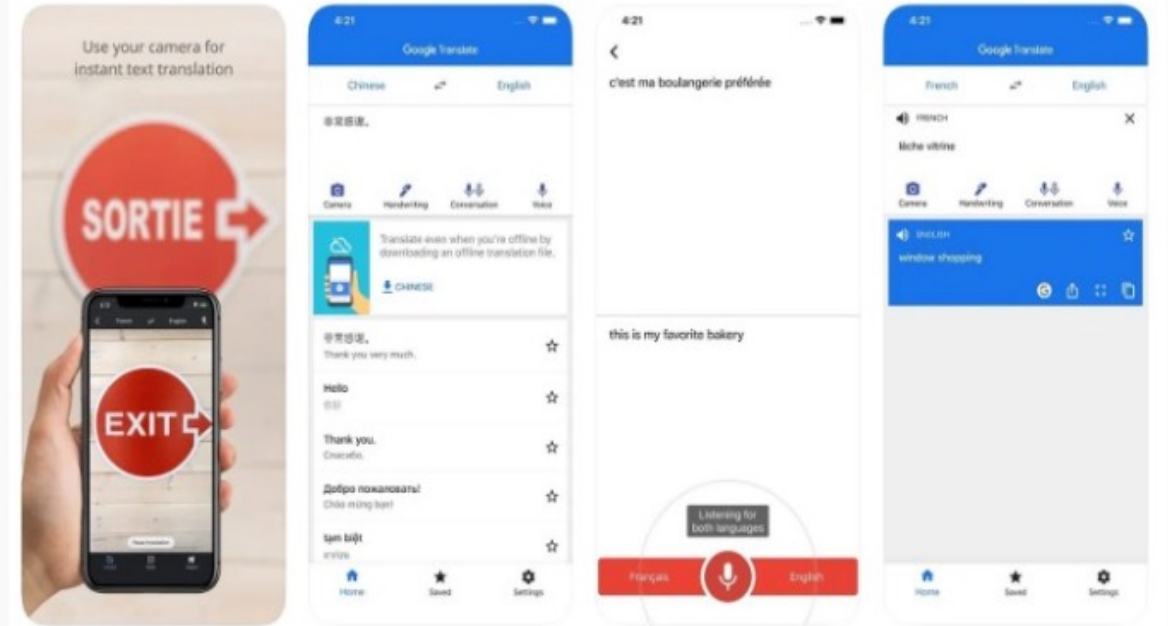


NLP IN RETAIL:

NLP

in Industries

NLP and [Chatbot](#) in Retail



NLP

in Industries

NLP IN RETAIL:

LANGUAGE TRANSLATION:

- NLP-powered language translation apps help users in many ways. For instance, Google Translate app, iTranslate, TripLingo, etc
- Travel apps help users to communicate and improve traveling experience

NLP

in Industries

NLP IN RETAIL:

IN-STORE VIRTUAL ASSISTANTS:

- **Customer service:** Customers could ask Millie questions about individual products and upcoming sales. She could also explain product features to customers while they are examining them or trying them on.
- **Personalized customer experience:** Millie can purportedly discern the preferences of returning customers. The software may create customer profiles based on the customer's name and appearance, which may necessarily involve machine vision.
- **Inbound Marketing:** For example, Millie can purportedly pose for selfies with potential customers and collect contact information in exchange.

NLP

in Industries

NLP IN RETAIL:

CHAT-BASED PRODUCT RECOMMENDATIONS:

- **Product recommendations:** This purportedly enables the conversational interface to upsell customers, guide them through gift shopping, and answer product information questions more accurately.

ChatGPT



Examples

"Explain quantum computing in simple terms"

"Got any creative ideas for a 10 year old's birthday?"

"How do I make an HTTP request in Javascript?"



Capabilities

Remembers what user said earlier in the conversation

Allows user to provide follow-up corrections

Trained to decline inappropriate requests



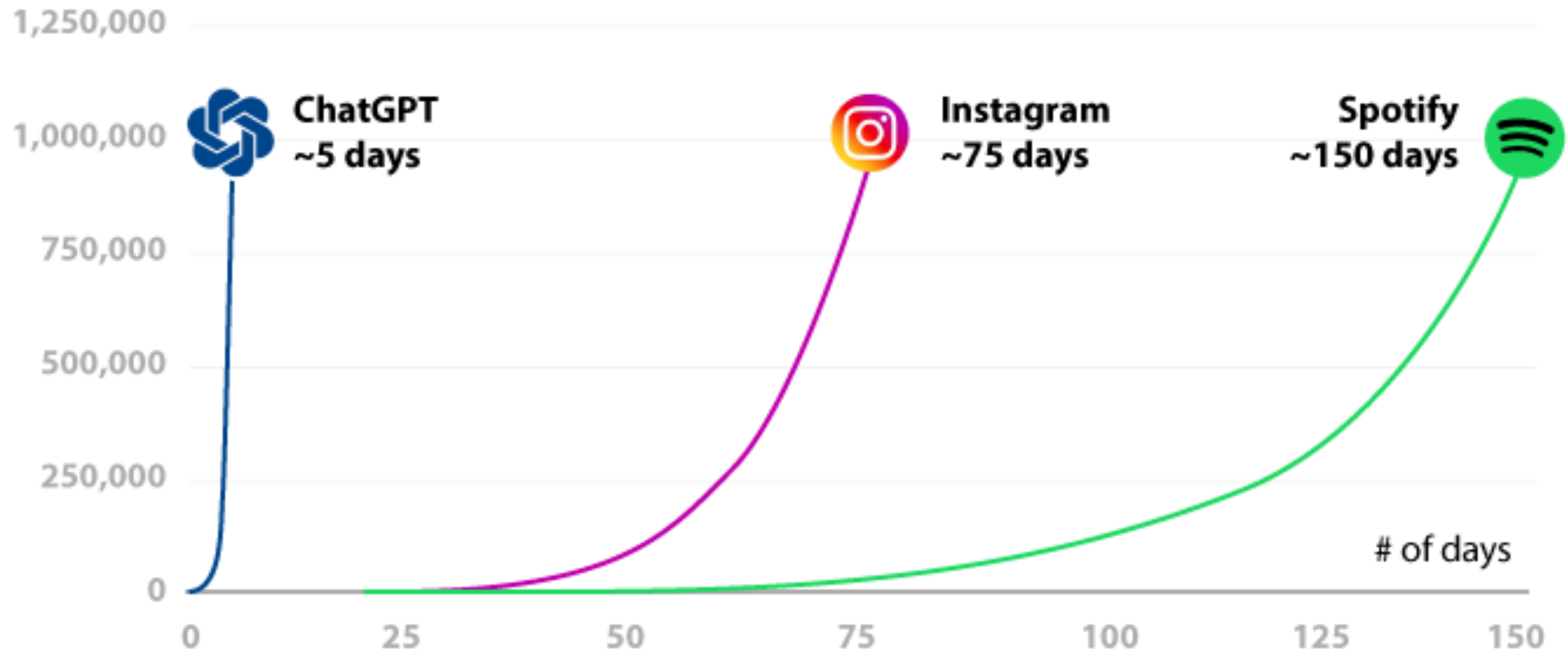
Limitations

May occasionally generate incorrect information

May occasionally produce harmful instructions or biased content

Limited knowledge of world and events after 2021

~ Path to 1 million users* (# of days from launch)



Sources: Google, Subredditstats, Media Reports

Thanks &
Credit

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